Introdução Amplificador Operacional

Amp-op

Eletrónica

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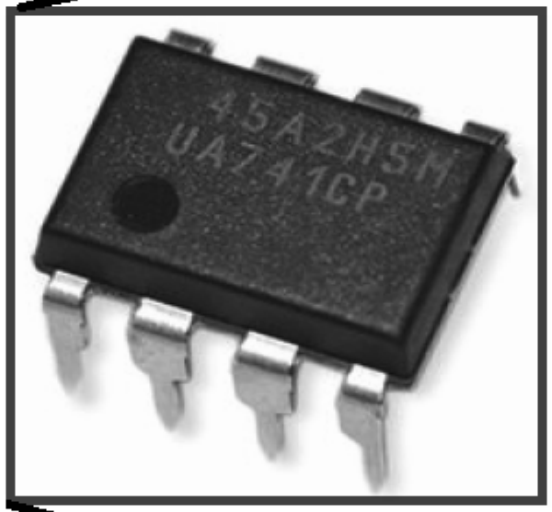
16/11/2022

[1] Acácio Amaral, "Eletrónica Aplicada", Edições Sílabo, 2021.

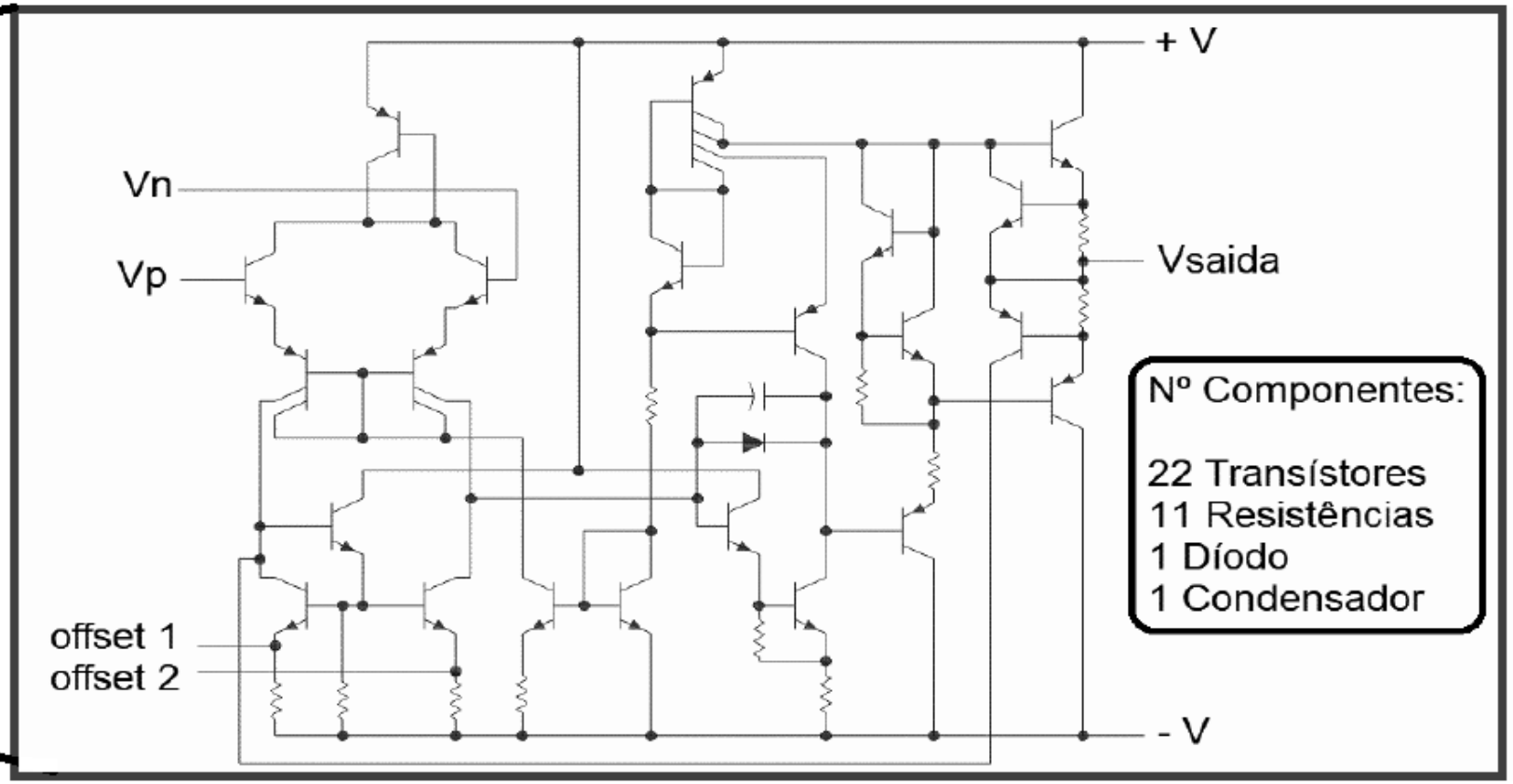
[10] https://www.ti.com/lit/ds/symlink/ua741.pdf

O Amplificador Operacional (Amp-Op) → é um CI que permite[1]:

- Amplificar sinais analógicos → Amp.
- Realizar diversas operações matemáticas sobre sinais analógicos $(+;\ -;\ \int f(x)dx;\ \frac{df(x)}{x})$ → Op.

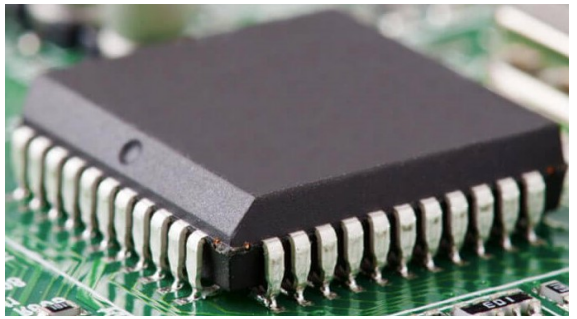


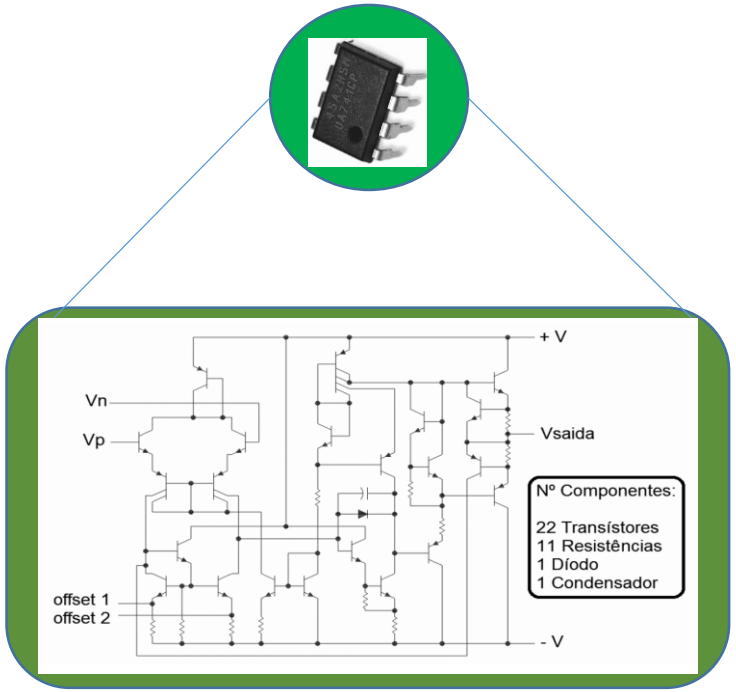
Possui vários andares com circuitos amplificadores

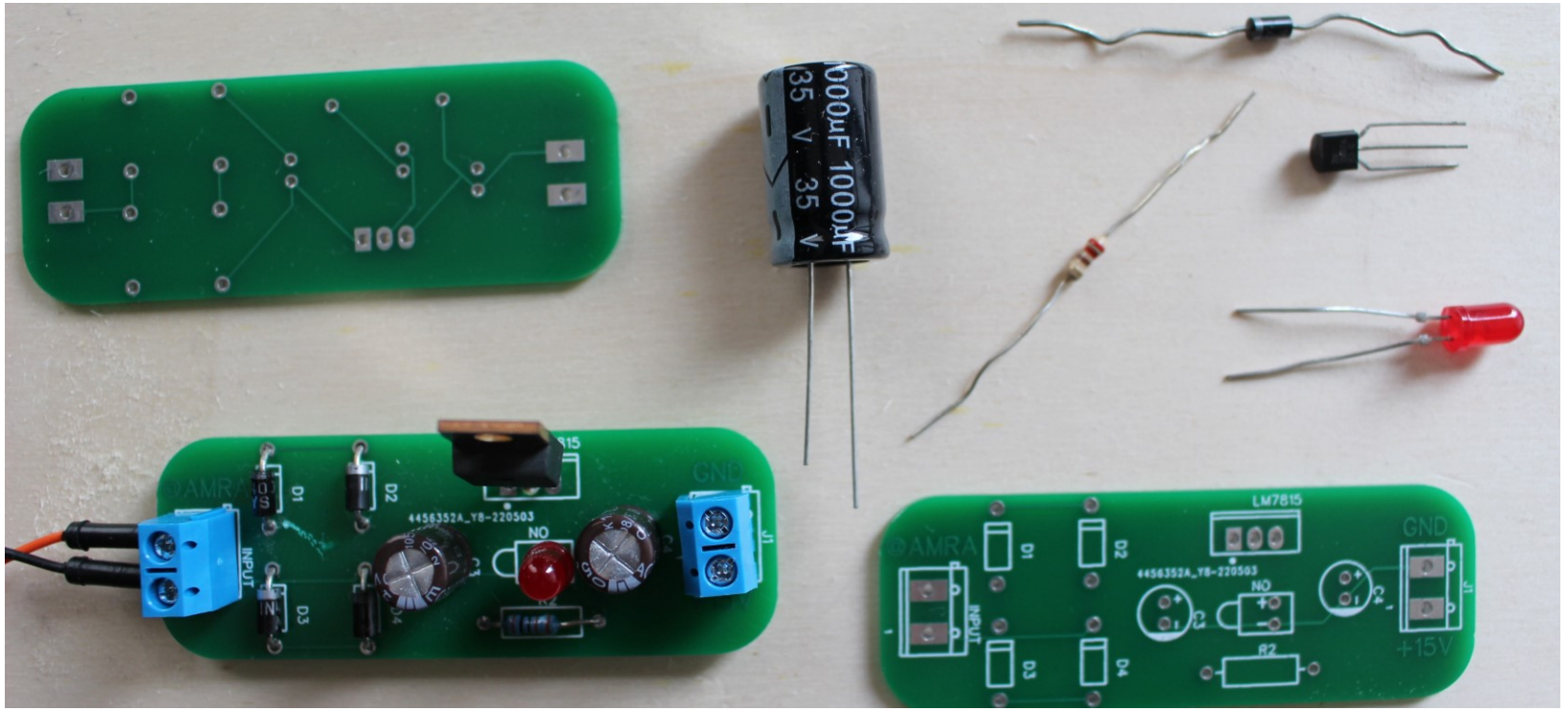


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- Circuitos integrados *versus* circuitos discretos[1].
  - Dimensão.
  - Topologia (elementos ativos *versus* elementos passivos).
  - Processamento de informação versus processamento de energia.







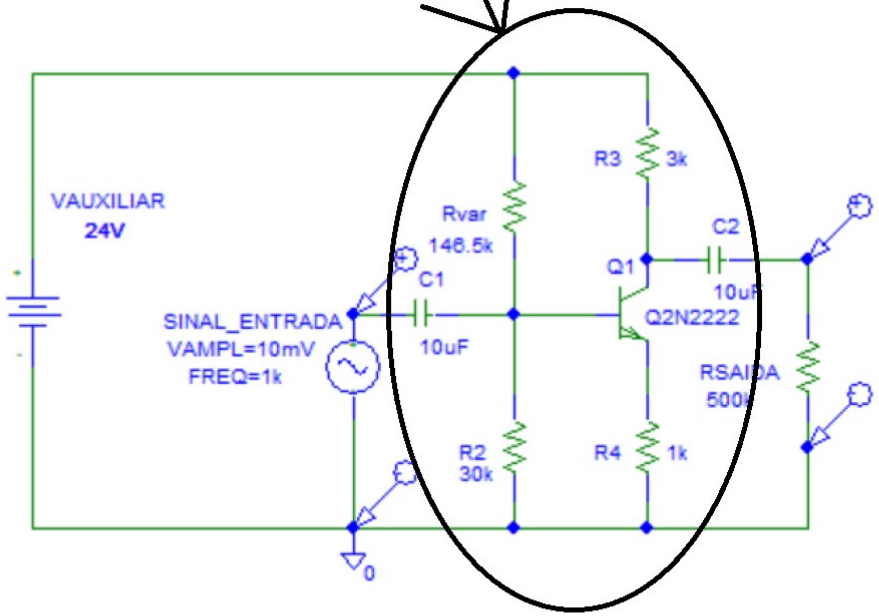
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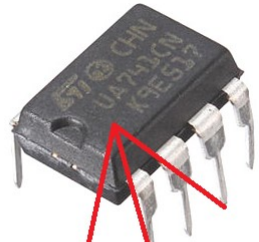
Circuitos Integrados

versus

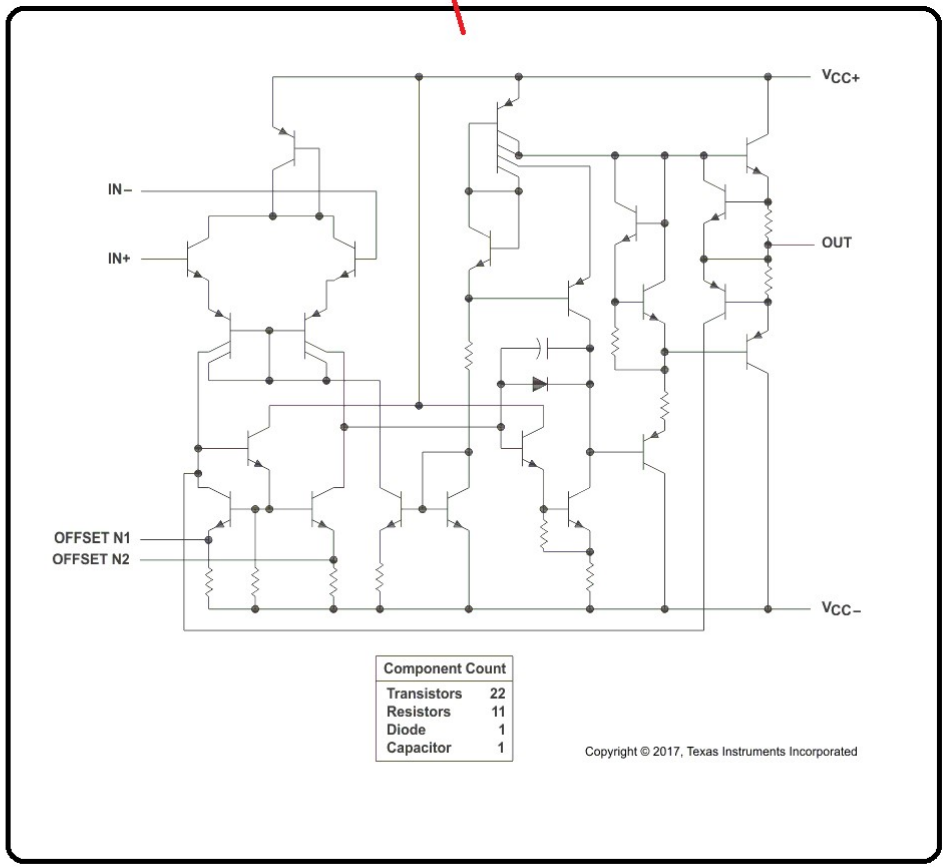
Circuitos Discretos (breadboard).







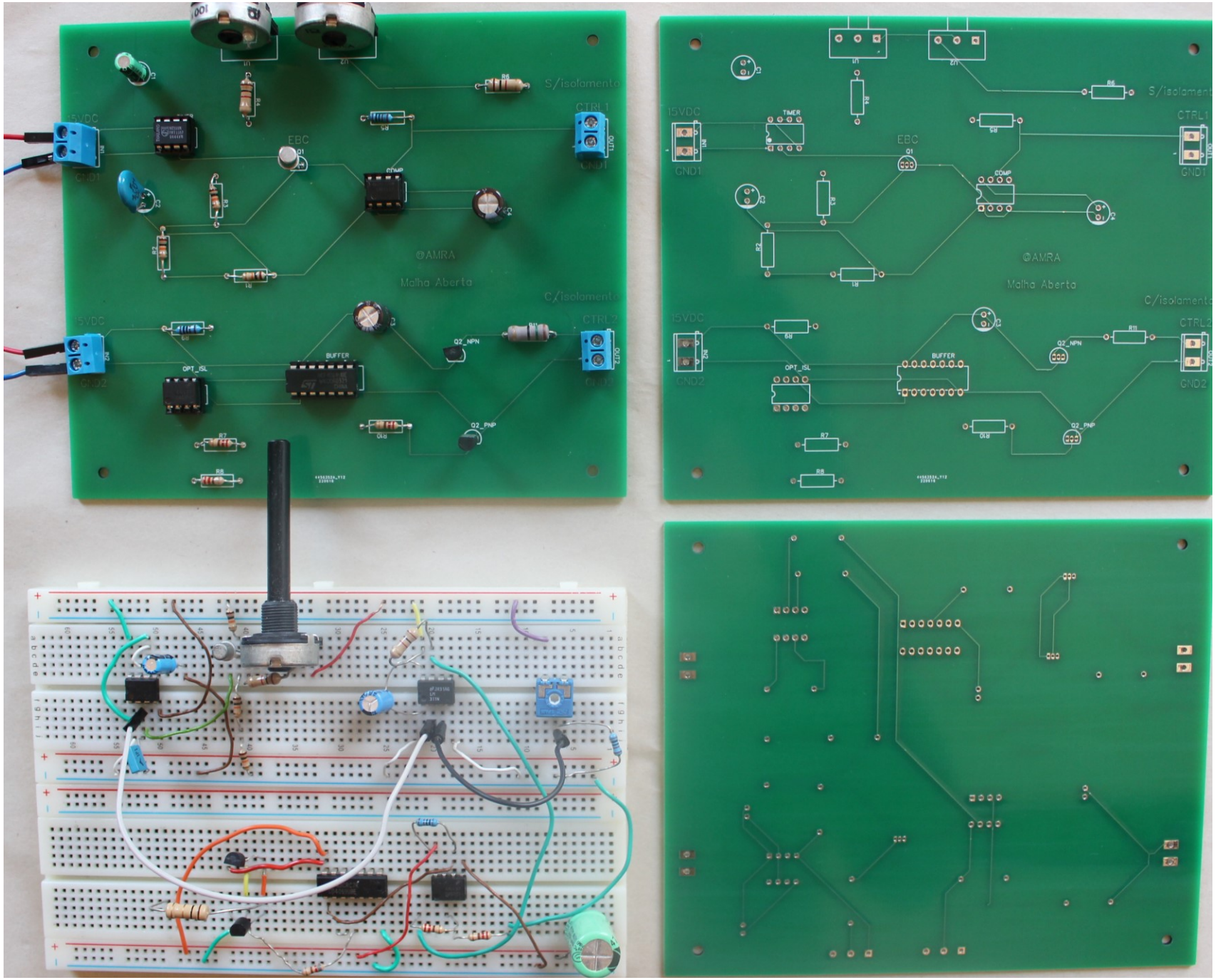
CI - Circuito integrado



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PCB → placa isolante revestida por fina película de cobre (fios condutores) revestidos por ligas (Estanho, chumbo), que representam o circuito onde são soldados os componentes eletrônicos.

Circuitos Híbridos
ex: Sistemas de Controlo
(Breadboard versus PCB)



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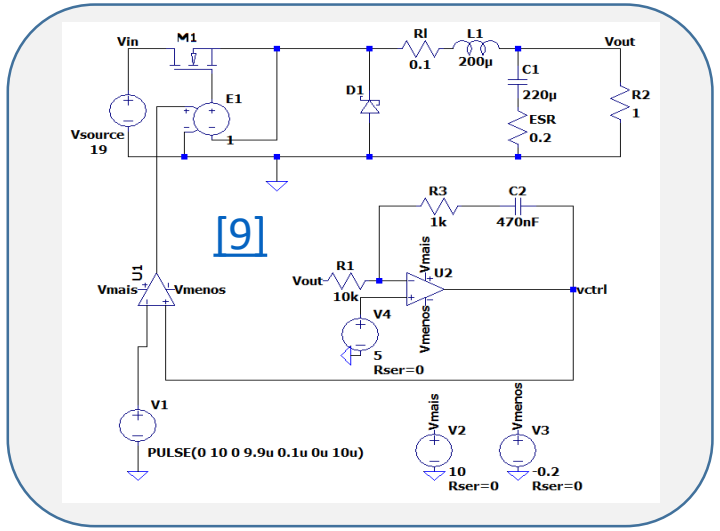
[1] Acácio Amaral, “Eletrónica Aplicada”, Edições Sílabo, 2021.

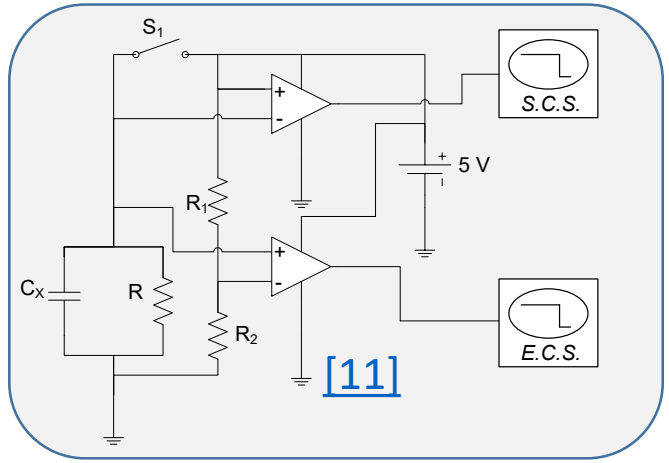
[11] Amaral, A. M. R., Cardoso, A. J. M. (July 2013). Using a Very Simple Capacimeter to Evaluate Aluminum Electrolytic Capacitors Health Status. International Journal on Engineering Applications, 4(1), pp. 234-240.

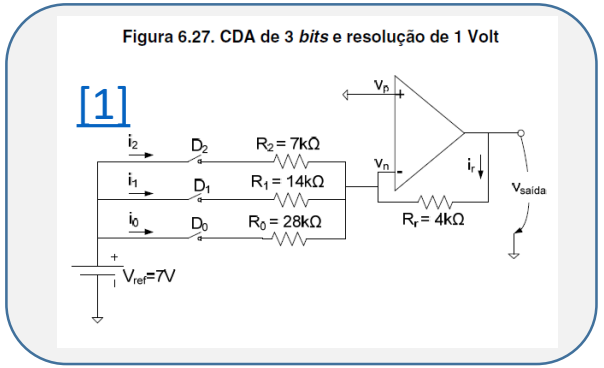
[9] Amaral, A. M. R., Cardoso, A. J. M. (Maio 2022). Using Python for the Simulation of a Closed Loop PI Controller for a Buck Converter. Signals, 3, pp. 313-325.

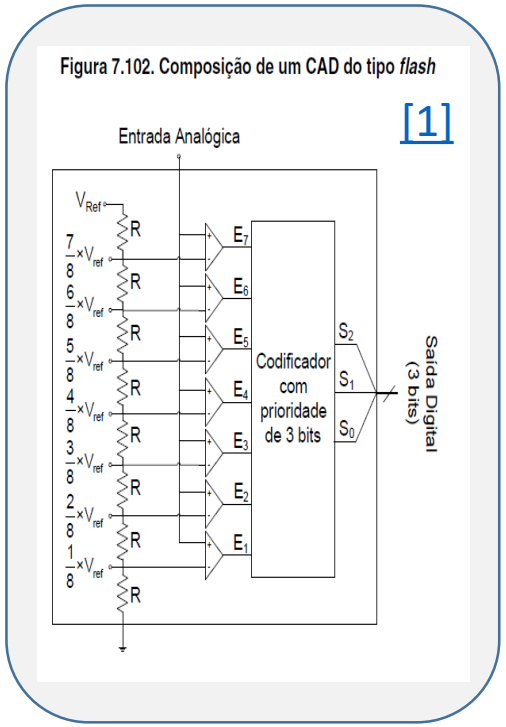
[12] https://spectrum.ieee.org/not-your-fathers-analog-computer

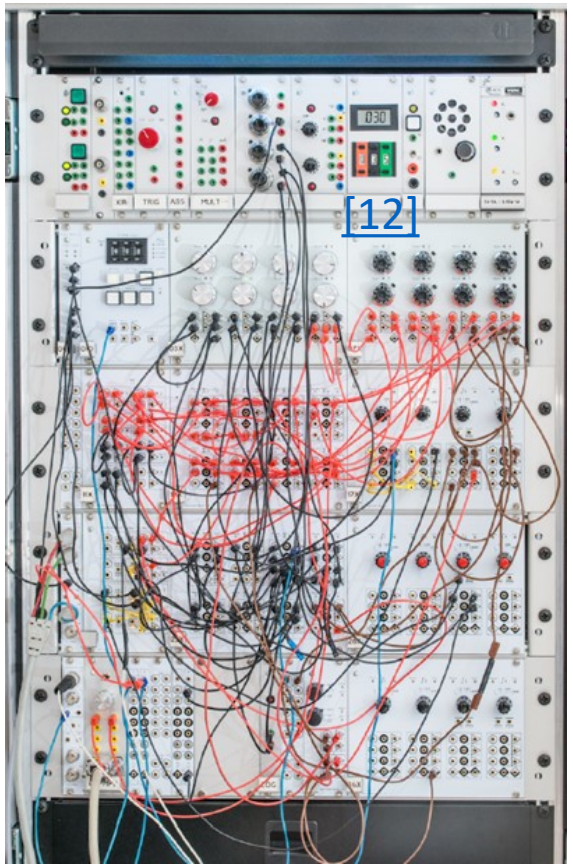
- Aplicações do Amp-op[1].
  - Computador Analógico.
  - Sistema de controlo[9]
  - Sistemas de medida[11].
  - CAD.
  - CDA.
  - Simulação de sistemas físicos (simples, barato, fiável, seguro e rápida conceção) .

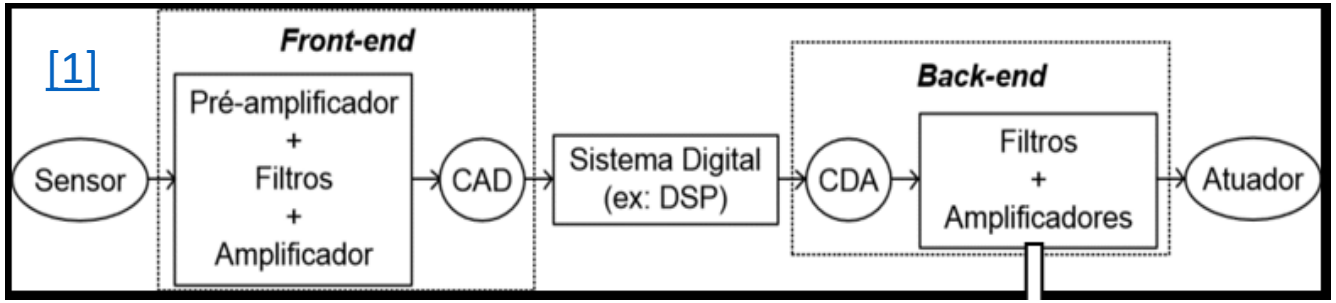












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Figura 7.26. Sensor de temperatura – LM35[30]

[1]

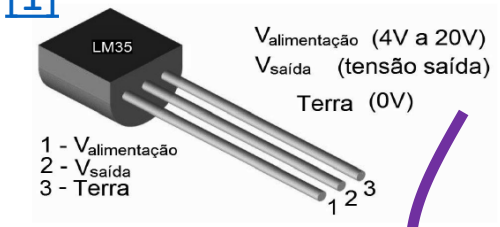
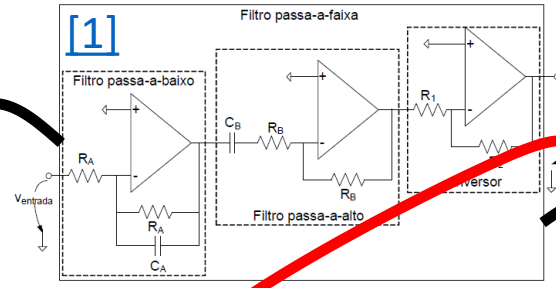
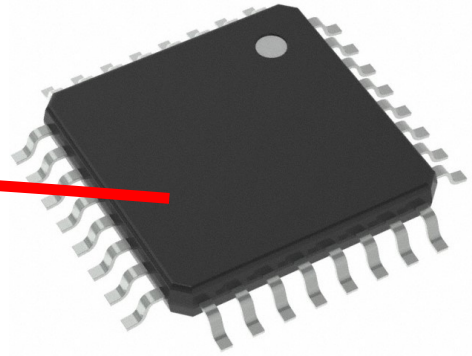


Figura 6.59. Filtro ativo do tipo passa-a-faixa

[1]





[1]

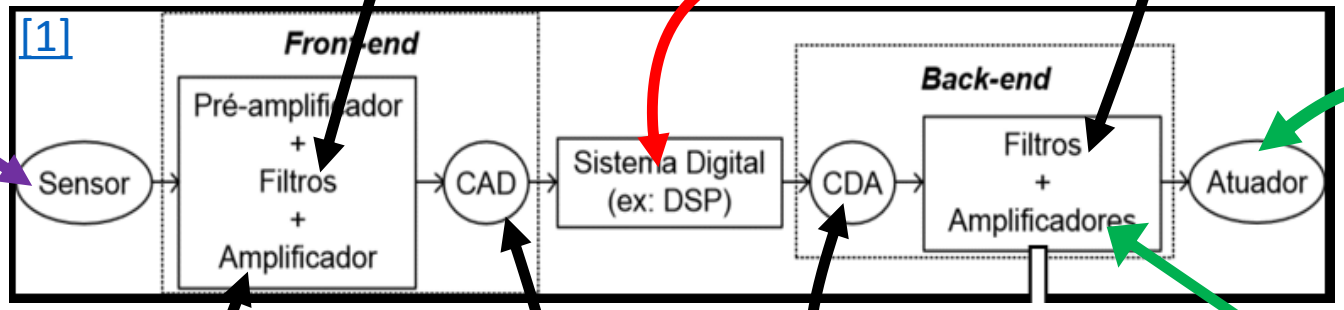




Figura 6.5. Circuito amplificador não inversor – cálculo da resistência de entrada em malha fechada

[1]

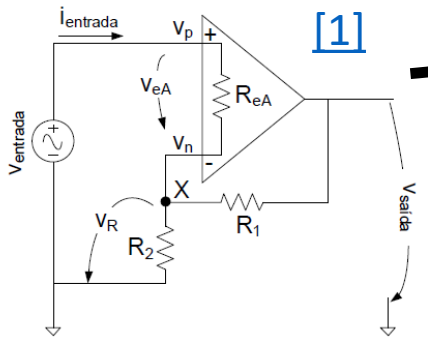


Figura 7.102. Composição de um CAD do tipo *flash*

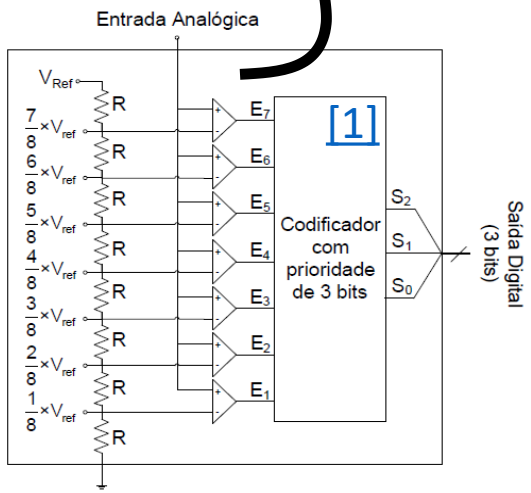


Figura 6.27. CDA de 3 *bits* e resolução de 1 Volt

[1]

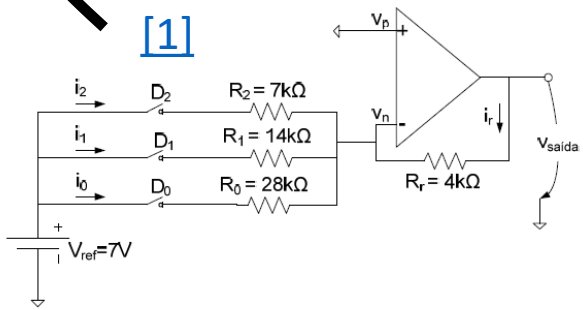
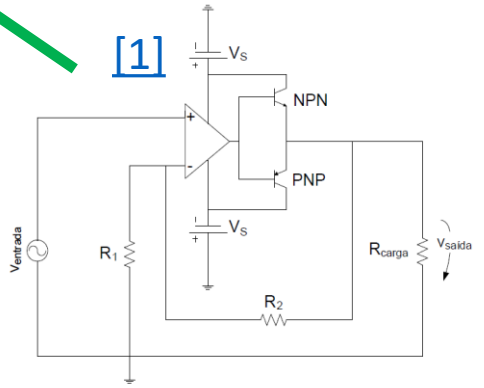


Figura 6.45. Amplificador classe *B* com um μA741 e dois transístores BJT interligados na configuração *push-pull*

[1]

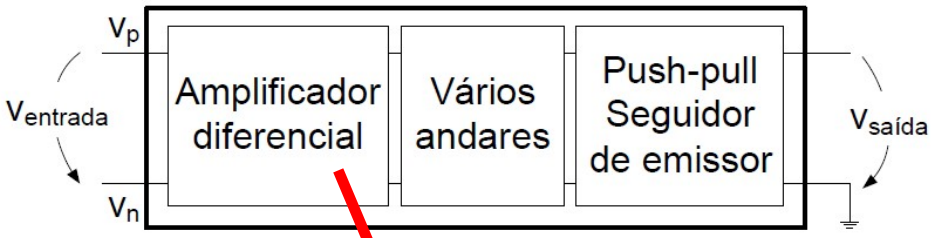


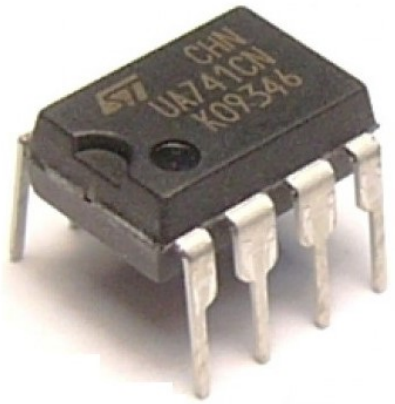
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- μA 741 → Padrão mínimo exigido pela indústria [1].

Figura 5.14. Diagrama de blocos do μA741

[1] →

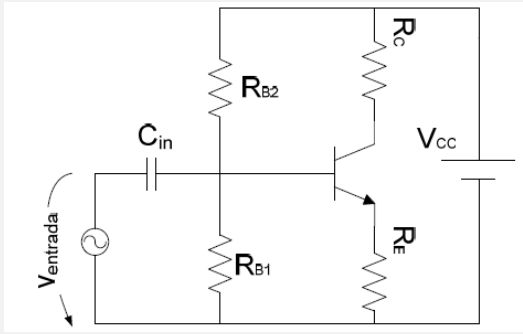




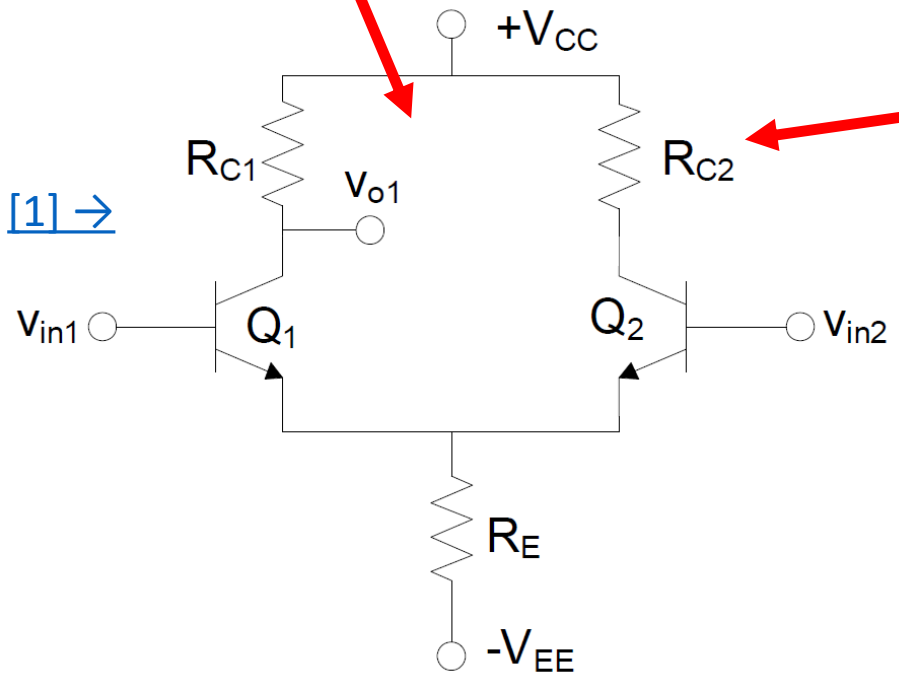
Amplificador em emissor comum:

$$Ganho = \frac{RC}{RE}$$

[1] →



[1] →



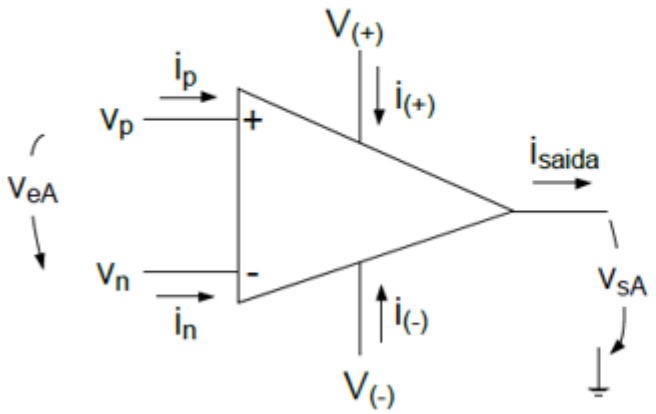
Amplificador diferencial (Características):

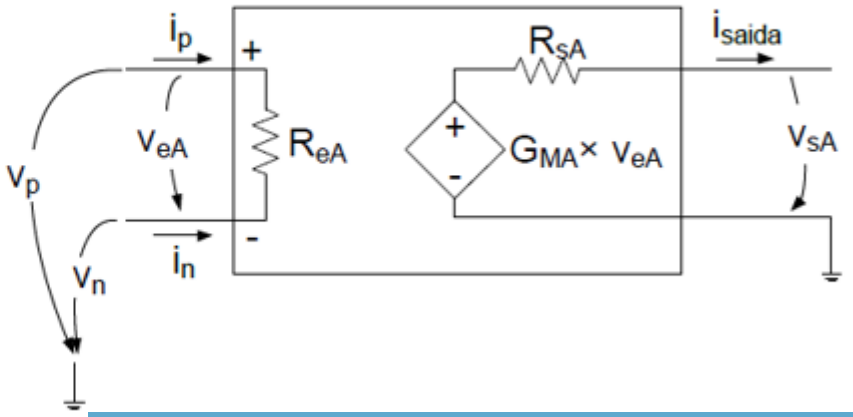
- Alimentação simétrica (+V_{CC}, 0 V, - V_{EE}).
- Imune ao ruido elétrico (FRMC ↑↑).
- Imune à variação da temperatura.

[1] → $$FRMC = \left| \frac{G_{Vd}}{G_{VMC}} \right| = \frac{\frac{R_C}{2 \times r_e}}{\frac{R_C}{r_e + R_E \times 2}} \cong \frac{R_E}{r_e}$$

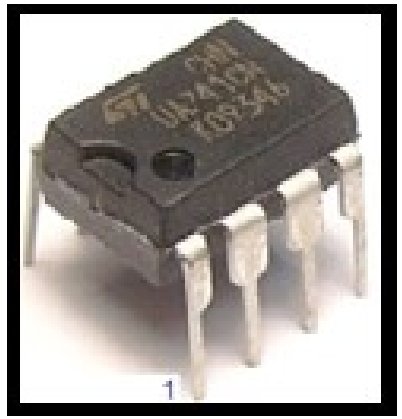
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Símbolo versus modelo matemático (Amp-op)

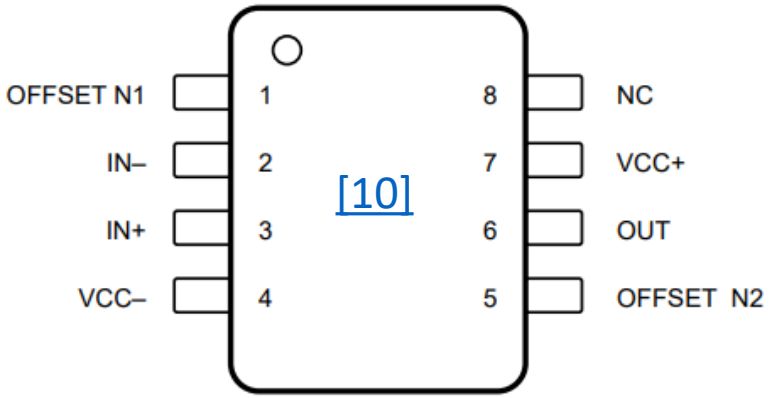




←[1]



uA741C D, P, or PS Package
8-Pin SOIC, PDIP, SO
Top View



Not to scale

[10]

Características Amp-op

Amp – op Ideal

$R_{entrada} = +\infty\ \Omega \mapsto R_{saída} = 0\ \Omega \mapsto G_{MA} = +\infty \mapsto$ Largura de banda $= +\infty$

Amp – op μA741

$R_{entrada} = 2\ M\Omega = +2 \times 10^6\ \Omega \mapsto R_{saída} = 75\ \Omega \mapsto G_{MA} = 10^5$

Largura de banda $= 1\ MHz = 10^6\ Hz$

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$$v_{sA} = \begin{cases} V_{(-)} & \text{se } G_{MA} \times (v_p - v_n) < V_{(-)} \\ G_{MA} \times (v_p - v_n) & \text{se } G_{MA} \times (v_p - v_n) \in [V_{(-)}, V_{(+)}] \\ V_{(+)} & \text{se } G_{MA} \times (v_p - v_n) > V_{(+)} \end{cases} \qquad (5.25)$$

←[1]

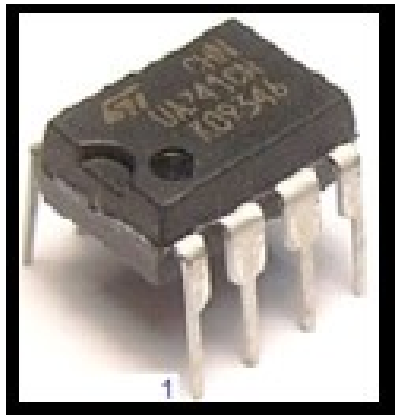
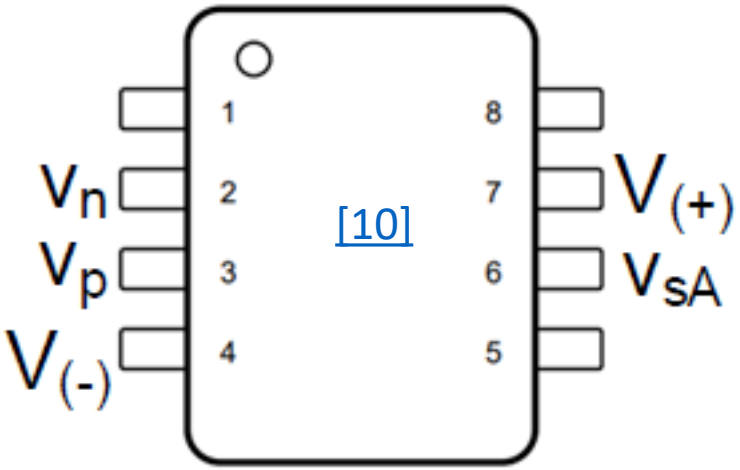
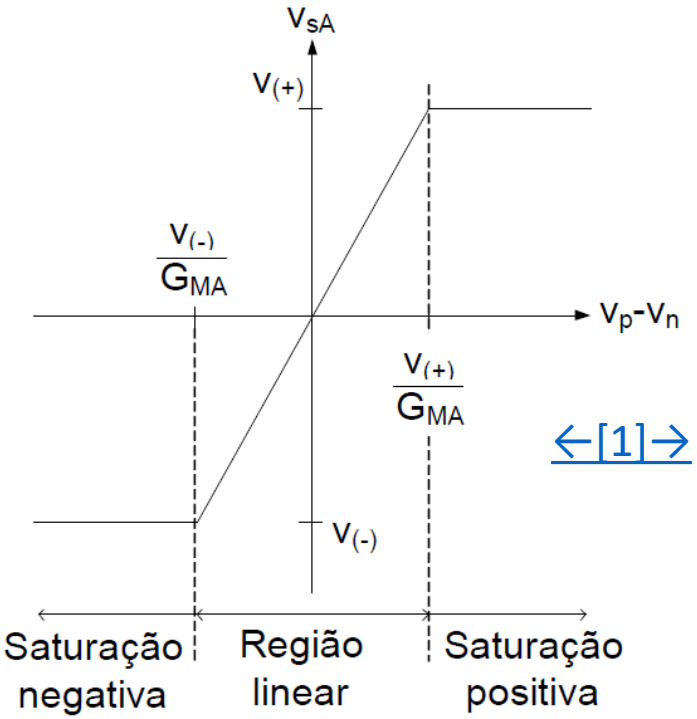
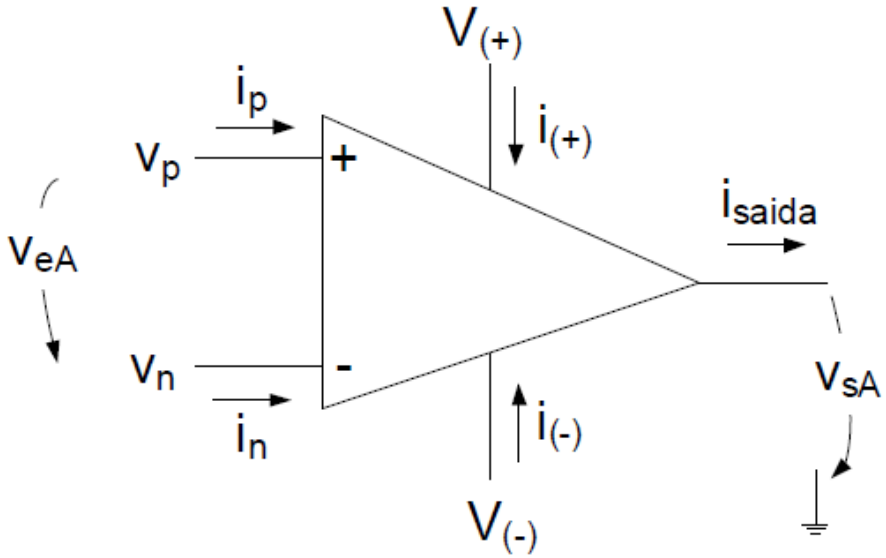


Figura 5.17. Função transferência do *amp-op*



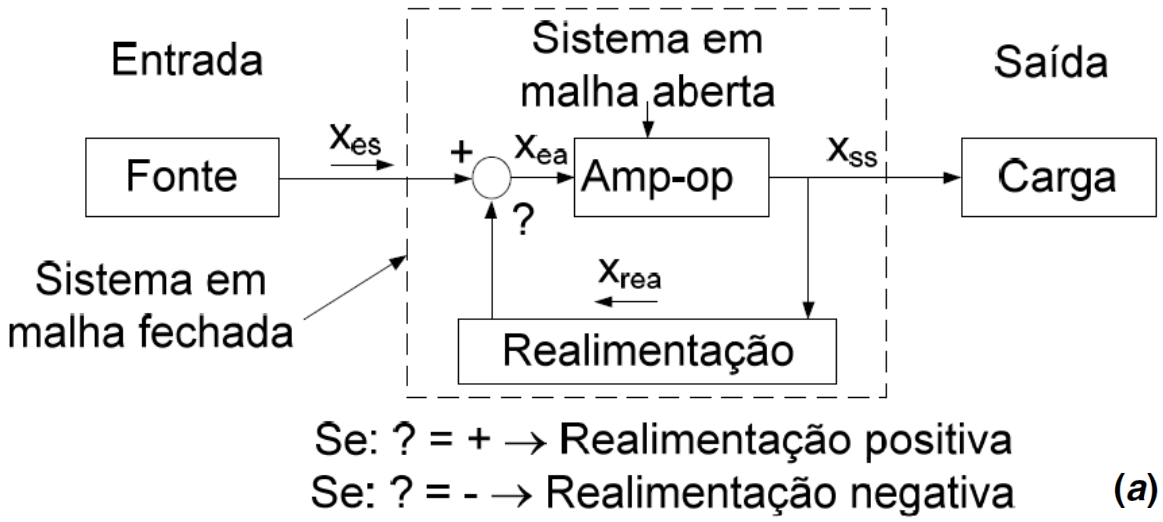


←[1]→



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Figura 5.18. Conceito de realimentação



Se: ? = + → Realimentação positiva
Se: ? = - → Realimentação negativa

←[1]

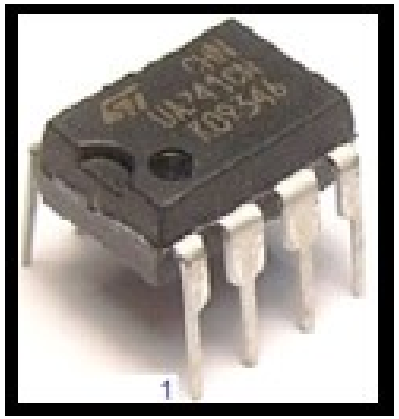
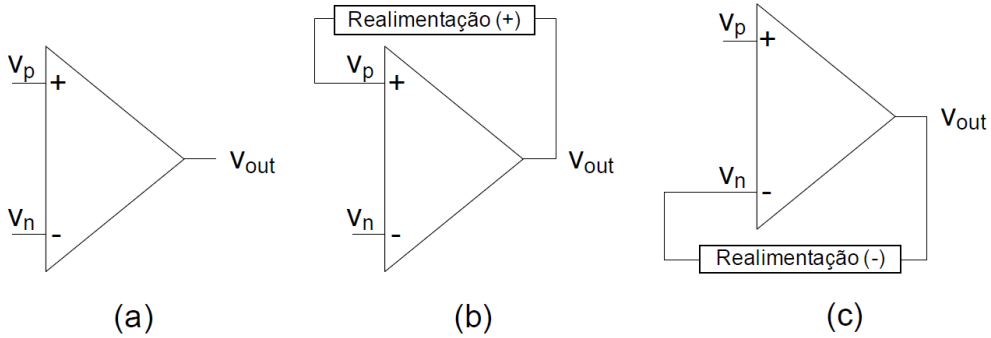


Figura 5.19. Modos de operação do *amp-op*:
(*a*) malha aberta, (*b*) realimentação positiva e (*c*) realimentação negativa

$$\begin{cases} G_{MF} = \dfrac{x_{ss}}{x_{es}} = \dfrac{G_{MA}}{1 + K \times G_{MA}} & \text{(R-)} \\ G_{MF} = \dfrac{x_{ss}}{x_{es}} = \dfrac{G_{MA}}{1 - K \times G_{MA}} & \text{(R+)} \end{cases} \quad (5.30)$$

←[1]→



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Na realimentação negativa

$$G_{MA} = 10^5$$

condição de curto-circuito virtual

$$(v_p - v_n) \cong 0 \quad \Rightarrow \quad v_p \cong v_n$$

$$R_{entrada} = 2\,\mathrm{M}\Omega = +2 \times 10^6\ \Omega$$

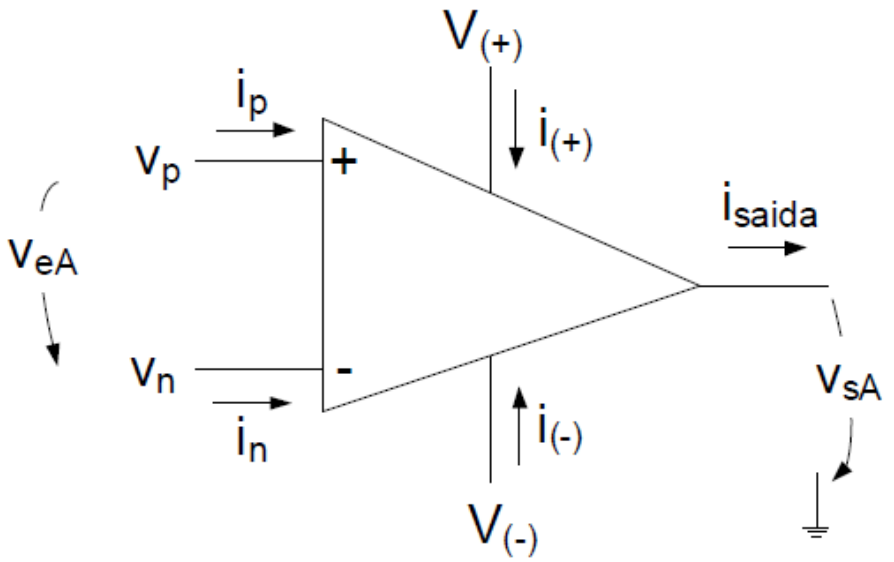
$$i_p \cong i_n \cong 0$$

←[1]→

condições na análise de circuitos

$v_p \cong v_n$.

$i_p \cong i_n \cong 0$.



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